

R² Value for different Algorithms for the dataset of 50_Startups:

1.Multi Linear Regression : R_score = 0.9358680970046241

2.Support Vector Machine:

S.No	C Value (Hyperparameter)	R_Score				
		Linear	Poly	rbf	sigmoid	precomputed
1	1.0	-0.0557	-0.0571	-0.05741	-0.0572	-(Error due to matric requirement)
2	10	-0.0396	-0.05366	-0.05680	-0.0547	”
3	100	0.1065	-0.0198	-0.0507260	-0.0305	”
4	500	0.5929	0.1147	-0.024323	0.0706	”
5	1000	0.7803	0.2662	0.00676	0.1851	”
6	2000	0.8768	0.4810	0.0675	0.3971	”
7	3000	0.8957	0.63700	0.1232	0.5914	”

The good SVM model is the one with the parameters of linear and 3000 c value (hyperparameter) as it has higher rscore comparatively

3. Decision Tree:

S.No	Criterion	Max_features	Splitter	R_Score
1	squared_error	sqrt	best	0.5758
2	squared_error	sqrt	random	0.1905
3	squared_error	log2	best	0.7657
4	squared_error	log2	random	0.3794
5	squared_error	None	best	0.9237
6	squared_error	None	random	0.9282
7	friedman_mse	sqrt	best	0.6178
8	friedman_mse	sqrt	random	0.5626
9	friedman_mse	log2	best	0.7627
10	friedman_mse	log2	random	0.0099
11	friedman_mse	None	best	0.8930
12	friedman_mse	None	random	0.8527
13	absolute_error	sqrt	best	0.5448
14	absolute_error	sqrt	random	0.6239
15	absolute_error	log2	best	0.9088
16	absolute_error	log2	random	0.8603
17	absolute_error	None	best	0.9706
18	absolute_error	None	random	0.6683

19	poisson	sqrt	best	0.2717
20	poisson	sqrt	random	0.0859
21	poisson	log2	best	0.4526
22	poisson	log2	random	0.1323
23	poisson	None	best	0.9318
24	poisson	None	random	0.8805

The optimal decision tree model is configured with the absolute_error criterion, max_features=None, and splitter=best, since it obtained the highest R² score among the evaluated models.

4.Randomforest Rscore

S.No	Criterion	Max_features	R_Score
1	squared_error	sqrt	0.7901
2	squared_error	log2	0.8064
3	squared_error	None	0.9395
4	friedman_mse	sqrt	0.8389
5	friedman_mse	log2	0.7999
6	friedman_mse	None	0.9318
7	absolute_error	sqrt	0.8062
8	absolute_error	log2	0.8068
9	absolute_error	None	0.9508
10	poisson	sqrt	0.7798
11	poisson	log2	0.7753
12	poisson	None	0.9303

The optimal random forestmodel is configured with the absolute_error criterion and max_features=None, since it obtained the highest R² score among the evaluated models.